

Challenge tackled

A parent whose child has an untreated rare disease ends up running the research effort: finding similar diseases, existing studies, and people who can help. That knowledge is spread over a dozen databases. We built a knowledge graph of **22 rare neurological diseases** and a website that takes a patient-group leader from a disease name to related diseases, reusable studies, collaborators and a cited next step. Every link shows its source, and gaps are stated rather than filled.

Tools and models

OpenAI gpt-oss-20b / 120b (via Groq)	Reads PubMed abstracts and proposes facts; the larger model reviews each fact; it also words the next steps in plain language.
Python, pandas, pyarrow	Joins ten public sources (MONDO, HPO, Orphadata, HGNC, Reactome, Gene Ontology, ClinVar, PubMed, ClinicalTrials.gov, NIH RePORTER) into two parquet files.
networkx	Disease similarity (information-weighted overlap of symptoms and pathways) and Louvain clusters.
FastAPI	Read-only JSON API over the graph; also serves the site.
React, Framer Motion, canvas, d3-geo	The website. The graph and the world map are drawn by hand on canvas.

public sources → **build** (join, extract with gpt-oss, verify, review, score) → **graph files** → API → website

What worked well

- **Evidence on everything.** About 4,000 facts, each with source, date and strength. A check script fails the build if one is missing.
- **LLM proposes, code verifies.** Of 460 facts gpt-oss proposed, 40 were dropped because the quote was not in the abstract. The reviewer then caught real errors, such as a gene attached to the wrong Batten subtype.
- **Clusters by mechanism, not by name.** The four Sanfilippo types land together through the heparan sulfate pathway.
- **Beyond the brief.** 1,618 study sites in 54 countries on a world map, milestones toward a treatment, and why earlier trials stopped.
- **Runs with one command**, no keys.

What was challenging

- **LLM quota.** The free tier allows 200,000 tokens a day per model. We lowered reasoning effort (about half the tokens per abstract), alternated two models, and spread reading across diseases. Still only 120 of 712 abstracts were read; the site says so.
- **Names that collide.** CLN1–CLN8 are also yeast genes; "Batten disease" means seven diseases. We filter on context and never pin a family-level gene statement on one subtype.
- **A better signal that made things worse.** Adding Gene Ontology to the similarity score split a real cluster, so we use it for navigation only.
- **Ranking people.** Counting diseases per researcher put someone whose six studies were withdrawn or terminated first. Listed as a known limit.

How we spent our time

Hours 0–2	Disease list, biology and research layers, similarity and clusters, first gpt-oss extraction.
Hours 2–5	First app in Streamlit; UI redesign in parallel (intro animation, staged graph); stricter quote checks.
Hours 6–8	Backend rebuilt around an API: provider fallback, second-model review, patient organisations read from their own websites, Gene Ontology.
Hours 8–10	React site: constellation graph, evidence drawer, tabbed disease page, Landscape tab and world map.
Last hours	Documentation for evaluators, fresh-clone test, videos and submission.

Dataset: in the repository under data/ (described in data/README.md). **If we had 24 more hours**, we would read the remaining 590 abstracts on a paid tier, add contradiction cases to the graph, and let a patient group add a disease from the website.